



Prices versus quantities in effort managed fisheries

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August 5, 2026



Outline

1. Problem statement



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4. Two Policy Approaches: Prices vs quantities



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12. Results



Problem statement

Using the standard Gordon-Schaefer model, if we introduce a license fee on fishing effort so that we raise the cost of fishing at what level should we set the fee? What would the demand for fishing licenses be? How would exogenous shocks like climate change impact license fee revenues?



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- ▶ If we restrict effort what fee should be charged that is consistent with the level of fishing effort?
- ▶ Should we tax landings, catch, effort or something else?
- ▶ What impact might climate change have on license fee revenues?



Literature

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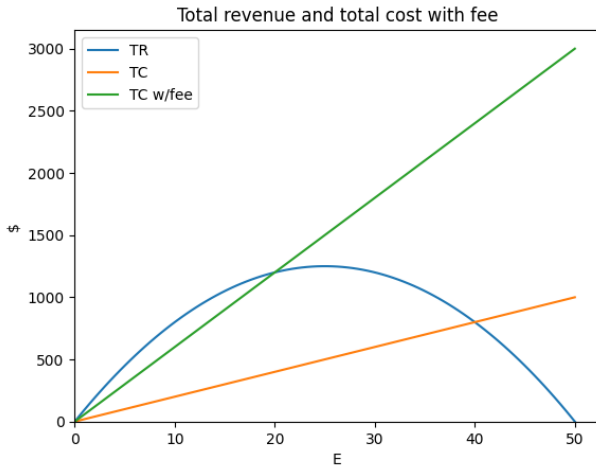


Two policy approaches: Prices vs quantities

- ▶ Firstly, only issue licenses equivalent to the the number of vessels corresponding to maximum economic yield E_{MEY} .
- ▶ Alternatively charge a license fee that raise the cost of fishing to the point where vessels will purchase licenses up to E_{MEY} .



Gordon-Schaefer Model



Optimal License Fee

$$f^* = \frac{1}{2}(pqK - c) \quad (1)$$



Proof

Rent is $pqK(1 - \frac{q}{r}E_{mey})E_{MEY} - c_fE_{MEY} = 0$, note that E_{mey} cancels on both sides. Set revenue per unit effort equal to $c_f = c + f$ so that rent is equal to zero at E_{MEY} . Inserting $E_{MEY} = \frac{r(Kpq-c)}{2Kpq^2}$ we get $pqK(1 - (q/r)\frac{r(Kpq-c)}{2Kpq^2}) = c + f$, r and Kpq^2 cancel to get $Kpq - \frac{1}{2}Kpq + \frac{c}{2} - c = f$ or $f^* = \frac{1}{2}(pqK - c)$.

Note: The argument here is fundamentally different to that made by Warming. See also link to the Colab notebook at the end of my paper



Equivalence results (Revenue equivalence)

Small island developing states often don't have domestic fleets. So rent maximization as a policy goal doesn't make much sense. Why would a country want to maximize benefits to foreigners? An alternative to this is to maximize license fee revenue.



Equivalence results

- ▶ Maximizing rent is equivalent to maximizing license fee revenues
- ▶ The maximand in both cases is either, or the fee f , assuming you know the demand for licenses
- ▶ The latter can be derived from the model by varying f .



Two-part pricing

The licensing and access fee problem is actually a two-part pricing problem.

- ▶ License fee is the variable component
- ▶ Access fees are fixed

$$\text{▶ } T = f^* E_{MEY} + CS = \frac{r(Kpq-c)^2}{2Kpq^2}$$

when comparing to the literature keep in mind this is a normative model, the literature tends to focus on either positive models or mixes up normative and positive models

Uncertainty quantification

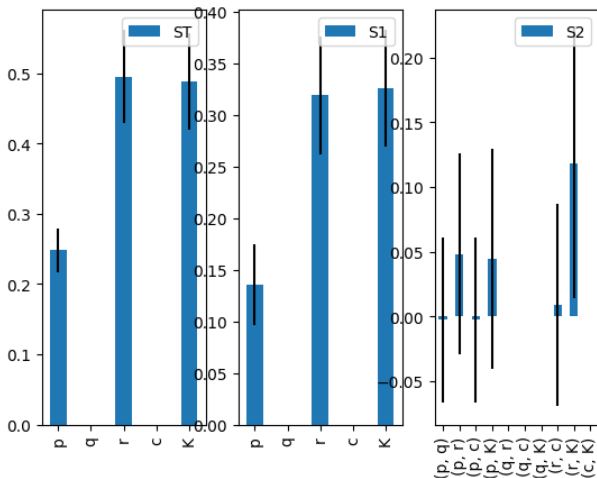
I use generalized sensitivity analysis aka variance based sensitivity analysis using the Sobol method. Essentially this is a Quasi-Monte-Carlo experiment that decomposes the variance of a model output (license fee revenues) into sub-components for each parameter. The table depicts the first order Sobol sensitivity indices and the total sensitivity indices

	$S1$	$S1_{conf}$	ST	ST_{conf}
p	0.13575885909376686	0.0	0.2	0.0
q	6.197573559165204e-08	0.0	0.0	0.0
r	0.31906702756744276	0.1	0.5	0.1
c	5.152078980462031e-07	0.0	0.0	0.0
K	0.3258437200243	0.1	0.5	0.1



Box plots

First and second-order Sobol sensitivity indices





Conclusion

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Conclusion

- ▶ An optimal license fee that implements maximum economic yield
- ▶ The demand for licenses
- ▶ Revenue equivalence between maximizing rent and maximizing licensing revenue is shown.
- ▶ Revenue equivalence between maximized license fee revenue and consumer surplus in the market for licenses is shown.
- ▶ Local and Global sensitivity analyses of the impact of key parameters on license fee revenue are shown



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Thanks for listening!

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